1.1.46

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The Question

Consider the AC bridge shown below. A, R, L, and C are positive finite values. The input is $v_{in}(t) = A\sin(\omega t)$. Find the condition for $V_0 = 0$.

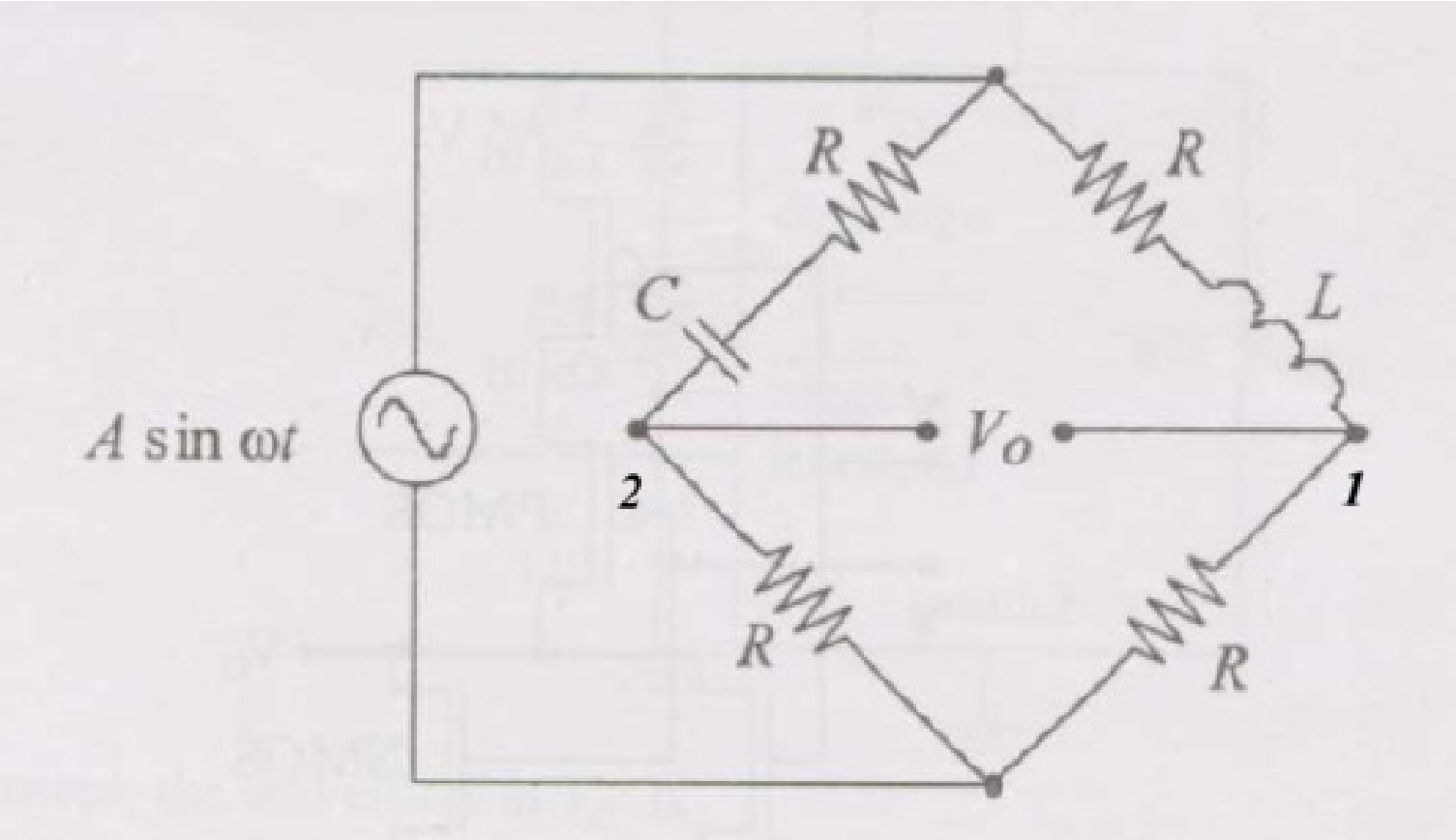

Figure: Circuit Diagram

Theoretical Solution: Laplace Domain

The transfer function is defined as $H(s) = \frac{V_0(s)}{V_{in}(s)}$. Applying the voltage divider rule:

$$V_1(s) = V_{in}(s)\left(\frac{R}{R + (R + sL)}\right) = V_{in}(s)\left(\frac{R}{2R + sL}\right) \tag{1}$$

$$V_2(s) = V_{in}(s)\left(\frac{R}{R + \left(R + \frac{1}{sC}\right)}\right) = V_{in}(s)\left(\frac{sRC}{2sRC + 1}\right) \tag{2}$$

The output voltage is $V_0(s) = V_1(s) - V_2(s)$. Thus:

$$H(s) = \frac{R}{2R + sL} - \frac{sRC}{2sRC + 1} \tag{3}$$

$$H(s) = \frac{R\,(2sRC + 1) - sRC\,(2R + sL)}{(2R + sL)\,(2sRC + 1)} \tag{4}$$

$$H(s) = \frac{R\left(1 - s^2 LC\right)}{(2R + sL)\,(2sRC + 1)} \tag{5}$$

Steady-State Analysis

For $V_0 = 0$ in steady-state, we set $H(j\omega) = 0$:

$$1 - s^2 LC = 0 \implies s^2 = \frac{1}{LC} \tag{6}$$

Substituting $s = j\omega$ for sinusoidal steady-state:

$$(j\omega)^2 = \frac{1}{LC} \tag{7}$$

$$-\omega^2 = \frac{1}{LC} \implies \omega^2 LC = -1 \tag{8}$$

Since $\omega, L, C > 0$, the product $\omega^2 LC$ is always positive.

Conclusion:
No real frequency ω can satisfy $\omega^2 LC = -1$. Therefore, V_0 cannot be zero.

Complete Python Implementation

```python
import numpy as np
from scipy import signal
import matplotlib.pyplot as plt

R, L, C = 100, 0.1, 100e-6
A, f = 10, 50
w = 2 * np.pi * f

# Transfer Function: H(s) = (R - s^2RLC) / (2s^2RLC + s(L + 4R^2C) + 2R)
num = [-R*L*C, 0, R]
den = [2*L*R*C, L + 4*R**2*C, 2*R]
sys = signal.TransferFunction(num, den)

t = np.linspace(0, 0.2, 200)
vin = A * np.sin(w * t)

tout, v0, _ = signal.lsim(sys, vin, t)

plt.plot(t, vin, label='Vin(t)', alpha=0.5, linestyle='--')
plt.plot(t, v0, label='V0(t)', linewidth=2)
plt.axhline(0, color='black', linewidth=2)
plt.axvline(0, color='black', linewidth=2)
plt.title('Time Domain Analysis of AC Bridge')
plt.xlabel('Time (s)')
plt.ylabel('Voltage (V)')
plt.legend()
plt.grid(True)
plt.savefig('plot.jpg')
plt.show()
```

Simulation Results

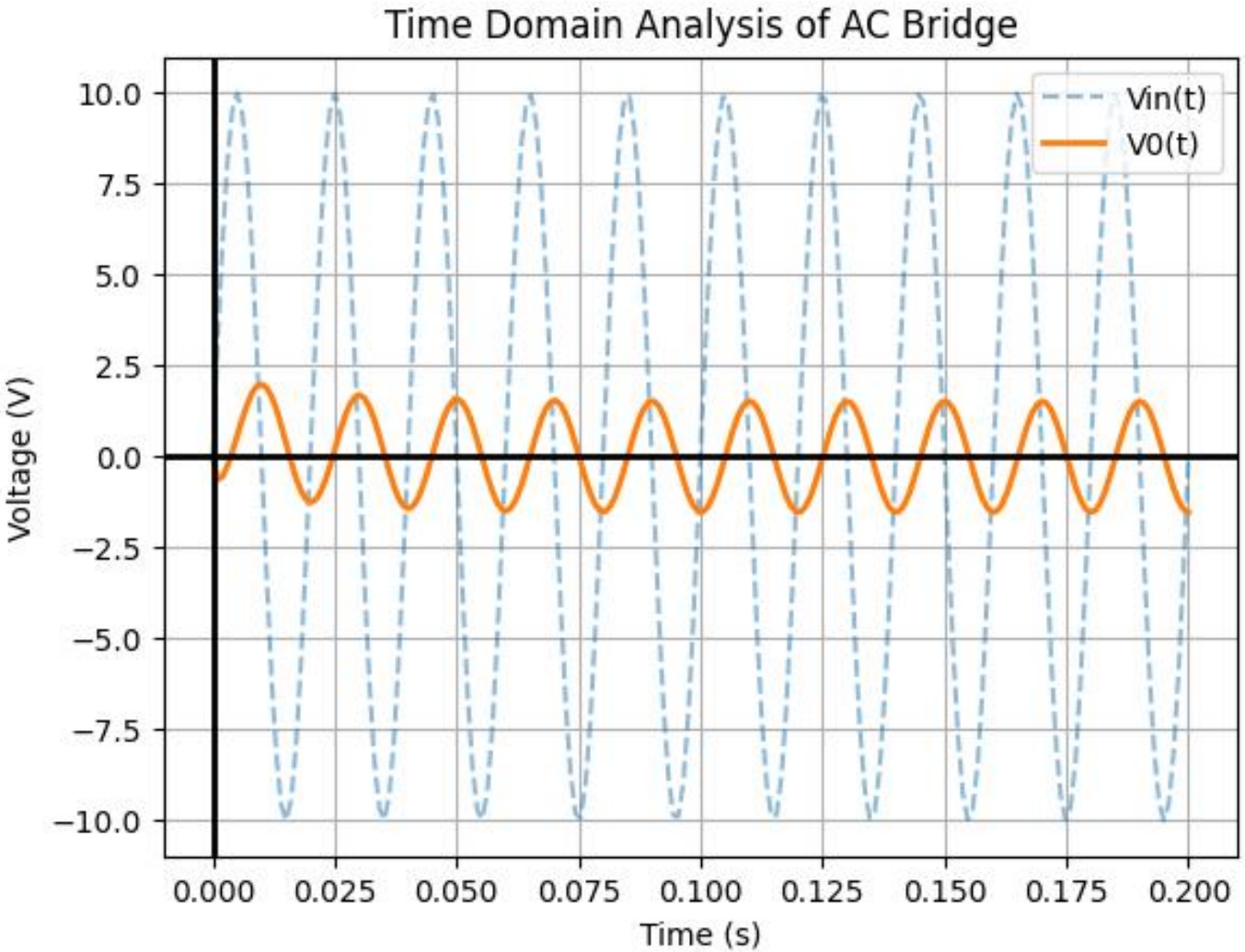

Figure: Time-Domain Response of V_{in} and V_0